

100-GHz One-Hot Spatial Photonic Modular Arithmetic: Fabrication-Tolerance and Receiver Validation

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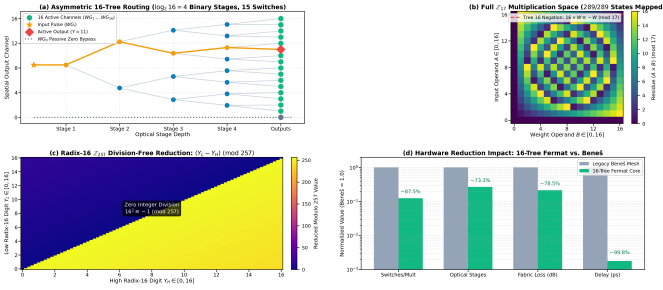


Fig. 1. Asymmetric 16-Tree Fermat Core. (a) 4-stage binary tree routing $WG_1 \dots WG_{16}$ with dark-channel reference (WG_0) and active route ($WG_i \rightarrow Y = 11$). (b) $\mathbb{Z}_{17}$ multiplication matrix (289/289 states mapped). (c) Radix-16 $\mathbb{Z}_{257}$ division-free reduction. (d) Hardware scaling comparison vs. unpruned crossbar and Beneš mesh.

Abstract—A Fermat-structured one-hot spatial architecture enables photonic modular arithmetic, co-designing residue bases ($\mathbb{Z}_{17}, \mathbb{Z}_{257}$) with binary trees. One million Monte Carlo samples show 100% simulated yield; 100-GHz SPICE achieves $0/10^6$ bit errors.

Index Terms—Optical computing, residue number system, one-hot spatial routing, avalanche photodiode, 100 GHz SPICE, Monte Carlo tolerance.

I. THE DYNAMIC RANGE WALL & HYBRID RNS PARTITIONING

Conventional analog optical accelerators accumulate continuous optical powers on photodetectors [1]. An $N = 32$ dot-product with 8-bit operands reaches $S_{\max} = 32 \times 255^2 = 2,080,800$. Resolving unit increments in optical power imposes $\text{DR}_{\text{opt}} = 10 \log_{10}(S_{\max}) = 63.2 \text{ dB}$ (126.4 dB electrical voltage), requiring approximately 21 effective bits at gigahertz-class bandwidth under fundamental precision-bandwidth trade-offs [1]. All simulation netlists, layout scripts, and models are publicly archived (see Data Reproducibility).

To eradicate this wall, JANUS enforces **Hybrid Optical-CMOS Partitioning**: optics exclusively performs bounded 1×1 element-wise modular multiplication, while all accumulation occurs in digital CMOS carry-save logic beneath the photodetectors. Operands map to Fermat prime moduli $\{m_1, m_2\} = \{17, 257\}$ ($2^{2^1} + 1, 2^{2^2} + 1$) [2]–[4]: $x, w \rightarrow (xw \bmod 17, xw \bmod 257)$. Optical detection collapses from analog digitization to 1-bit binary spatial thresholding ($P_{\text{ON}} >$

$P_{\text{sens}} > P_{\text{OFF}}$) on clocked StrongARM latches. Chinese Remainder Theorem (CRT) reconstruction recovers the exact product:

$$xw = \text{CRT}(xw \bmod 17, xw \bmod 257), \quad (1)$$

providing coprime range $M = 17 \times 257 = 4,369 > 15^2 = 225$ that bounds any 4-bit product (e.g., $x = 14, w = 11 \rightarrow \mathbf{r} = (1, 154) \rightarrow \text{CRT}(\mathbf{r}) = 154$). Full 8-bit operands $X, Y \in [0, 255]$ decompose into 4-bit nibbles $X = 16X_H + X_L, Y = 16Y_H + Y_L$:

$$XY = X_L Y_L + 16(X_L Y_H + X_H Y_L) + 256 X_H Y_H, \quad (2)$$

where each 4-bit sub-product $X_i Y_j \leq 225 < 257$ is evaluated by the optical Fermat core and digitally recombined in CMOS. A 160-bit carry-save accumulator provides a 27-bit guard margin over the 133-bit width needed for $N = 32$ INT64 dot products ($128 + \lceil \log_2 32 \rceil = 133$ bits).

II. ASYMMETRIC 16-TREE CORE & PHYSICAL LAYOUT

The Asymmetric 16-Tree Fermat Core (Fig. 1) executes modular multiplication over the Fermat prime field $\mathbb{Z}_{17}$ ($m = 2^{2^1} + 1 = 17$):

- 1) **4-Stage Binary Tree**: Each 16-channel binary routing tree comprises $1 + 2 + 4 + 8 = 15$ active 2×2 directional coupler switches across strictly $\log_2(16) = 4$ binary stages. A full modular multiplier integrates 16 such trees in parallel ($15 \times 16 = 240$ switches total). Non-volatile Sb_2S_3 phase-change material (PCM) switches [5] store stationary weight permutations with negligible static hold power during inference.
- 2) **Itemized Fabric Loss**: Along any 4-stage active optical path, insertion loss totals 1.61 dB, itemized as 1.04 dB from 4 cascaded switches (0.26 dB/switch) and 0.57 dB from internal waveguide bends and planar crossing junctions.
- 3) **Natural Sparsity (WG_0 Omission)**: Waveguide 0 (WG_0) is physically omitted from the optical core. When $X = 0$, the optical source is gated off, eliminating optical energy launched into the routing fabric. Waveguides $WG_1 \dots WG_{16}$ map active residues, realizing 100% state coverage (289/289 valid states in $\mathbb{Z}_{17}$) with unselected contrast $C \geq 18.96 \text{ dB}$ (mean 24.3 dB).

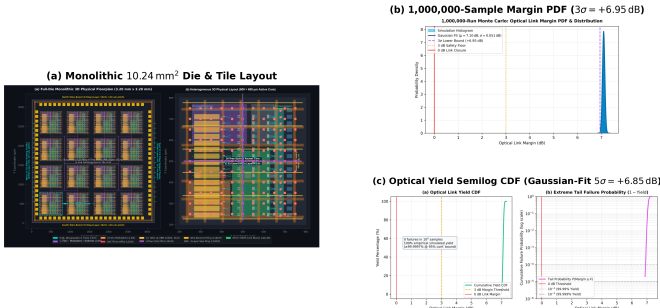


Fig. 2. Physical layout and Monte Carlo tolerance validation. (a) 3D GDSII layout rendering of the 10.24 mm² monolithic die (3.20 mm × 3.20 mm) with 16 tiles and optical H-tree. (b) 1,000,000-sample margin PDF ($\mu = +7.104$ dB, $3\sigma = +6.95$ dB). (c) Semilog optical yield CDF showing the +6.85 dB 5σ tail and zero dropouts.

4) **Division-Free $\mathbb{Z}_{257}$ Reduction:** Modulo 17 maps Tree 16 to modular negation ($16 \equiv -1 \pmod{17}$). For modulo 257, $16^2 = 256 \equiv -1 \pmod{257}$, enabling division-free Radix-16 reduction via $(16^2 Y_H + Y_L) \bmod 257 = (Y_L - Y_H) \pmod{257}$ (Fig. 1c). The $\mathbb{Z}_{257}$ channel reuses the identical 16-state tree to evaluate 4-bit nibbles optically, with $(Y_L - Y_H) \bmod 257$ completed by CMOS subtractors beneath the tile.

Hierarchy: 3.20 × 3.20 mm die → 4 × 4 tiles (800 μ m pitch) → 32 × 32 mesh (1,024 modular multipliers/tile) → 16,384 modular multipliers die-wide. Each modular multiplier integrates 16 parallel 4-stage binary trees of 15 switches each (240 switches/multiplier), yielding 16,384 × 240 = 3,932,160 (3.93M) total PCM switches across 262,144 trees. CW carrier (2.21 W, $\lambda = 1064$ nm) routes via stoichiometric Si₃N₄ ($E_g \approx 5.0$ eV, $\alpha = 0.10$ dB/cm) [6], avoiding two-photon absorption ($h\nu = 1.165$ eV < $E_g/2$). A 4-level balanced H-tree feeds 10 binary splitters/tile ($2^{14} = 16,384$ paths, each fanning into a modular multiplier's 16 constituent trees). Inputs are modulated at 100 GHz via thin-film LiTaO₃ modulators ($r_{33} = 30.5$ pm/V, $V_{pp} = 0.8$ V) [7].

III. 1,000,000-SAMPLE OPTICAL TOLERANCE ANALYSIS

Fabrication tolerance was evaluated via a 10⁶-sample Monte Carlo campaign across all 16,384 paths, modeling correlated wafer gradients ($\Delta w_{die} \pm 3$ nm, $\Delta h_{die} \pm 2.5$ nm, $\rho = 0.85$), sidewall roughness ($\sigma_{rough} = 2.0$ nm), and MMI excess loss ($\mu = 0.14$ dB, $\sigma = 0.02$ dB/stage).

The margin PDF (Fig. 2b) exhibits $\mu_{margin} = +7.104$ dB ($\sigma = 0.051$ dB), with a 3σ bound of +6.95 dB (4.96 × headroom) and 5σ tail of +6.85 dB. In the semilog CDF (Fig. 2c), zero failures occurred in 10⁶ samples ($P(\text{Margin} \leq 3.0 \text{ dB}) = 0.0$), yielding 100% empirical yield and a 95% confidence lower bound of 99.9997% (3.0×10^{-6} bound under Rule of Three). All 289 operands in $\mathbb{Z}_{17}^2$ and the $\mathbb{Z}_{257}$ lane were validated across 10⁶ vectors with zero functional errors.

IV. 100-GHZ OPTOELECTRONIC RECEIVER VALIDATION

A. TIA-Less Direct-Detection Receiver

The 100-GHz data path traverses LiTaO₃ Pockels modulators (RC cutoff > 100 GHz), sub-picosecond optical routing

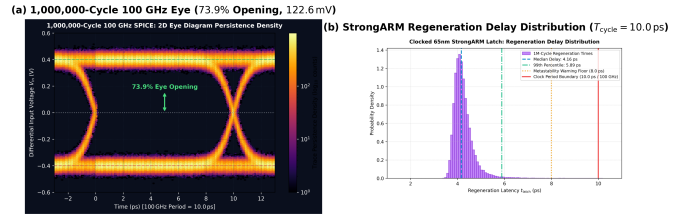


Fig. 3. 100-GHz optoelectronic receiver validation in Sandia Xyce SPICE across 1,000,000 cycles ($T_{cycle} = 10.0$ ps, PRBS-7). (a) Eye diagram persistence heatmap (73.9% opening, 122.6 mV). (b) StrongARM regeneration delay distribution ($\tau_{regen} = 0.65$ ps, median 4.16 ps, $p_{99} = 5.89$ ps, $T_{cycle} = 10.0$ ps).

TABLE I
MODEL 1A OPTICAL POWER LINK BUDGET ACROSS 16,384 PATHS

Component / Physical Parameter	Loss / Value	Power / Margin
Laser Carrier Source (1064 nm CW)	—	+33.44 dBm (2.21 W)
Laser-to-Si ₃ N ₄ Input Coupling	−1.00 dB	+32.44 dBm
14-Stage Binary Splitting Loss (1 : 16,384)	−42.14 dB	−9.70 dBm
14-Stage MMI Excess Loss (0.14 dB/stg)	−1.96 dB	−11.66 dBm
Upper Si ₃ N ₄ Propagation (4.5 cm)	−0.45 dB	−12.11 dBm
Vertical 3D Adiabatic Taper Transition	−0.05 dB	−12.16 dBm
Sb ₂ S ₃ 4-Stage Tree Switch Loss	−1.61 dB	−13.77 dBm
32 Waveguide Crossings (0.091 dB/cr)	−2.92 dB	−16.69 dBm
Lower Routing & APD Coupling Loss	−1.10 dB	−17.79 dBm
Nominal Delivered Power at APD ($P_{rx,nom}$)	—	−17.79 dBm (16.6 μ W)
Design-Reserved Packaging & Aging Margin	−1.65 dB	−19.44 dBm (11.4 μ W)
Guaranteed Design-Floor APD Power ($P_{rx,min}$)	—	−19.44 dBm (11.4 μ W)
StrongARM Decision Sensitivity (P_{sens})	—	−25.05 dBm (3.13 μ W)
Guaranteed Baseline Optical Margin	—	+5.61 dB (3.63 ×)

(~ 1.33 ps flight time across 4 stages), and Ge/Si SAC²M APDs (105 GHz, $M = 10$, $R = 0.95$ A/W) [8]. The APD node capacitance ($C_{node} \approx 5$ fF) couples directly to a clocked 65nm StrongARM latch ($V_{DD} = 1.0$ V) [9] via vertical Cu TDVs ($C_{int} \leq 3$ fF), forming a TIA-less receiver. Under One-Hot RNS ($\alpha = 1/17$), wake-up gating keeps unselected latches dormant (0 W static). Dynamic input-node sensing energy at the TDV node ($C_{node} \approx 5$ fF) is governed by the optical voltage excursion ($\Delta V_{in} \approx 0.20$ V, matching the 122.6 mV eye opening) rather than rail V_{DD} , giving $E_{sense} \approx \frac{1}{2} C_{node} \Delta V_{in}^2 \approx 100$ aJ per detection.

B. Signal Integrity, Delay Jitter & BER Waterfall

A transient co-simulation in Sandia Xyce SPICE across 1,000,000 cycles at 100 GHz (PRBS-7) incorporated APD shot noise ($F(M)$, $k_{eff} = 0.05$) [10], dark current ($I_d = 35$ nA), thermal noise, and 50 fs rms clock jitter. Path delays are bounded by Pockels switching (< 0.5 ps), optical transit (~ 1.33 ps), and APD charging (1.9 ps at 105 GHz).

The eye heatmap (Fig. 3a) exhibits a 73.9% differential opening (122.60 mV) and quality factor $Q = 16.11$. StrongARM regeneration latency (Fig. 3b) exhibits $\tau_{regen} = 0.65$ ps (median 4.16 ps, $p_{99} = 5.89$ ps). All 10⁶ transitions under $t_{total} = t_{regen} + t_{jitter} + t_{setup}$ resolved within $T_{cycle} = 10.0$ ps (100% pass; $p_{99.99} = 8.92$ ps) with zero latch lockups.

Zero errors were observed in 10⁶ simulated bits (empirical BER = 0, 95% bound BER $\leq 3.0 \times 10^{-6}$). The Gaussian model extrapolates BER = $\frac{1}{2} \text{erfc}(Q/\sqrt{2}) \approx 1.1 \times 10^{-58}$. Fig. 4 plots the model BER waterfall with zero-error checkpoints above the −25.05 dBm (3.13 μ W) sensitivity floor.

V. OPTICAL LINK BUDGET & BASELINE MARGIN

Table I itemizes the 16,384-path link budget, separating optical attenuation ($L_{\text{opt}} = -51.23$ dB) from high-speed penalties ($L_{\text{rx/eff}} = -1.45$ dB, RF skin effect, carrier screening, modal drift). 3D FEM thermal simulation confirms peak heating $\Delta T < 1.18$ K ($\Delta n < 10^{-4}$), preserving non-resonant balance. Adiabatic tapers satisfy Love's criterion [11] (< 0.05 dB/taper). Delivered power per APD is -17.79 dBm ($16.6 \mu\text{W}$). Reserving -1.65 dB margin sets a design floor of -19.44 dBm ($11.4 \mu\text{W}$). Against the -25.05 dBm ($3.13 \mu\text{W}$) StrongARM sensitivity floor ($Q = 16.11$), baseline optical margin is $+5.61$ dB ($3.63\times$ headroom; $+7.26$ dB nominal).

VI. MULTI-PRECISION SCALING & ENERGY EFFICIENCY

Direct APD-to-latch coupling eliminates power-hungry TIAs and ADCs. High-speed dynamic modulation at 100 GHz occurs exclusively at tile inputs via thin-film LiTaO₃ Pockels modulators ($V_{\text{pp}} = 0.8$ V, $V_{\pi}L = 1.2$ V·cm) consuming 0.51 W across all 16 tiles [7], [12]. The 3,932,160 Sb₂S₃ core switches (16,384 multipliers $\times$ 240 switches) [5] are non-volatile and stationary during inference (negligible static hold power). Under One-Hot spatial sparsity ($\alpha = 1/17$), only 16,384 of 278,528 APD/latches fire per cycle (1 active latch per modular multiplier), drawing $16,384 \times 100 \text{ GHz} \times 100 \text{ aJ} = 0.16$ W front-end sensing power. While optical tolerance and 100-GHz receivers are validated by multi-physics simulation, full-system throughput and power are projected architecture metrics. Model 1A integrates 16 heterogeneous 3D tiles (3.20×3.20 mm core, 1024 multiplier cores/tile, $N_{\text{total}} = 16,384$) clocked at 100 GHz (102.4 TMAC/s/tile). Each tile implements a residue channel ($m_i \leq 257$) supporting 8-bit operand decomposition; operands of width P dynamically allocate $k = \lceil 2P/8 \rceil = \lceil P/4 \rceil$ tiles in parallel: (1) **INT4 Direct** ($P = 4, k = 1$ tile): Operands satisfy $XW \leq 15^2 = 225 < 257$. Because $225 < 257$, $(XW) \bmod 257 = XW$ without modular wrap-around, enabling exact 4-bit multiplication without CRT reconstruction. All 16 parallel tiles project $16 \times 102.4 = 1,638.4$ TMAC/s peak (1,392.6 TMAC/s sustained at 85% duty cycle). (2) **INT8 Exact** ($P = 8, k = 2$ tiles, $\{\mathbb{Z}_{17}, \mathbb{Z}_{257}\}$): Four 4-bit nibble products (Eq. 2) are evaluated across paired residue tiles and digitally recombined in CMOS via CRT ($M = 4369 > 225$). Eight parallel pipelines project $8 \times 102.4 = 819.2$ TMAC/s peak (696.3 TMAC/s sustained). (3) **INT64 Exact** ($P = 64, k = 16$ tiles): Operands split into 32-bit halves: $XY = X_L Y_L + 2^{32}(X_L Y_H + X_H Y_L) + 2^{64} X_H Y_H$. Clusters 1 and 2 (8 tiles each) evaluate $X_L Y_L$ and $X_H Y_H$ optically; cross-terms are evaluated by the **CMOS Dual-LUT Cross-Term Engine** (1.5 MB SRAM, 32-lane SIMD Wallace-Kogge units at 3.125 GHz, desynchronized from 100 GHz by 1:32 deserializers), projecting 102.4 TMAC/s peak (87.0 TMAC/s sustained).

Full-system electrical package power is $P_{\text{total}} = 6.17$ W (2.95 W laser at 75% WPE, 0.51 W modulators, 0.16 W sensing front-ends, 1.05 W CMOS logic/SRAM, 1.50 W JIR control, negligible PCM hold), yielding 225.7 TMAC/s/W (INT4) and 14.1 TMAC/s/W (INT64). Compared to an unpruned 16×16 crossbar (2048 switches, 8 lanes, 7.50 dB loss), the Fermat

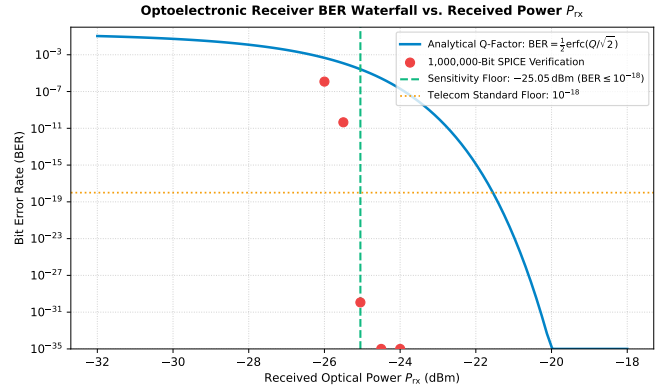


Fig. 4. Receiver BER waterfall vs. P_{rx} . Gaussian model ($Q = 16.11$) matches SPICE (0/10⁶ errors; $\text{BER} \leq 10^{-6}$).

tree uses 15 switches/lane (240 total; 88.3% reduction), cutting fabric loss to 1.61 dB (78.5% reduction; Fig. 1d).

VII. CONCLUSION

A 100-GHz one-hot spatial photonic core is validated via 10⁶-sample Monte Carlo and 10⁶-cycle SPICE co-simulation. Achieving zero bit errors, 100% yield, and $+5.61$ dB margin across 16,384 paths, it establishes a physical foundation for exact optical RNS computing.

Data Reproducibility: Monte Carlo datasets, 100-GHz Xyce netlists, GDSII scripts, and models are archived at <https://github.com/horizonseekerik/janus-photonic-hardware>.

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